

FormBook Alert Investigation

Network Traffic Analysis Exercise | Wireshark

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Exercise source	Malware-Traffic-Analysis.net - 2026-08-09 “First to Last” training exercise
Objective	Identify the compromised Windows client and investigate network activity associated with FormBook C2 alerts.
Primary tool	Wireshark
Scope	Packet-capture analysis only; no host forensics, payload execution, or malware reverse engineering performed.

Executive Summary

A packet-capture investigation was performed after simulated SOC alerts identified FormBook command-and-control (C2) activity. Analysis linked the suspicious traffic to internal host 172.16.8.49. Network and Windows-related protocols were then correlated to identify the endpoint as DESKTOP-5NLV63K, associated with account rvance / Raymond Vance. The client initiated TCP connections to alert-associated external IP addresses and generated suspicious HTTP GET and POST activity. The POST included a 909-byte form-urlencoded body containing a long encoded-looking value. The exact contents and full impact could not be determined from the PCAP alone.

Key Findings

Artifact	Observed value
Compromised client IP	172.16.8.49
MAC address	00:12:f0:28:d4:34
Hostname	DESKTOP-5NLV63K
User account	rvance
Full name	Raymond Vance
Alert-associated external IPs	172.64.155.76; 146.59.71.167
Observed HTTP artifacts	GET /ujvq/?...; POST /irpw/
POST body	909 bytes; application/x-www-form-urlencoded; begins with 2kn1=

Investigation & Evidence

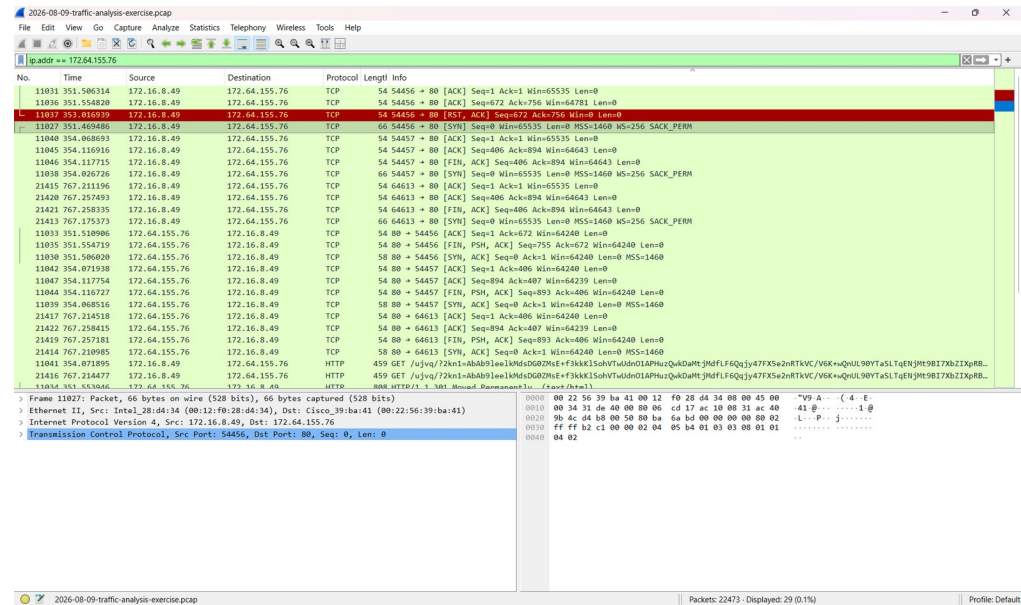


Figure 1. Traffic between 172.16.8.49 and alert-associated IP 172.64.155.76.

The alert-associated external address was used as an initial pivot. Traffic showed 172.16.8.49 communicating with 172.64.155.76, identifying the internal endpoint for further analysis.

No.	Time	Source	Destination	Protocol	Length	Info
1884	41.066530	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<00>
1899	41.828122	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<00>
2129	42.583534	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<00>
2477	43.346637	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<00>
1897	41.580882	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<20>
1900	42.339506	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<20>
2296	43.104491	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<20>
2489	43.863288	172.16.8.49	172.16.8.255	NBNS	110	Registration NB DESKTOP-5NLV63K<20>
1885	41.067624	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<00>
1898	41.827862	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<00>
2130	42.583663	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<00>
2476	43.346132	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<00>
3124	59.900710	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<1d>
11180	369.926555	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<1d>
19834	679.966462	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<1d>
2484	43.500075	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<1e>
2494	44.262293	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<1e>
2575	45.204915	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<1e>
2605	46.025951	172.16.8.49	172.16.8.255	NBNS	110	Registration NB FIRSTTOLAST<1e>
3125	59.900952	172.16.8.8	172.16.8.49	NBNS	104	Registration response, Name is owned by another node NB 172.16.8.8
11181	369.926973	172.16.8.8	172.16.8.49	NBNS	104	Registration response, Name is owned by another node NB 172.16.8.8
19835	679.966653	172.16.8.8	172.16.8.49	NBNS	104	Registration response, Name is owned by another node NB 172.16.8.8

Figure 2. NBNS registrations from 172.16.8.49 identify hostname DESKTOP-5NLV63K.

NBNS traffic linked the compromised IP to hostname DESKTOP-5NLV63K. Additional protocol analysis was used to attribute the endpoint to a user account.

2026-08-09-traffic-analysis-exercise.pcap

File Edit View Go Capture Statistics Telephony Wireless Tools Help

ip.addr == 172.16.8.49 && samr

No.	Time	Source	Destination	Protocol	Length	Info
2961	55.134071	172.16.8.8	172.16.8.49	SAMR	218	Close response
2965	55.135175	172.16.8.8	172.16.8.49	SAMR	218	Close response
2965	55.120350	172.16.8.49	172.16.8.8	SAMR	332	Connect5 request
2984	55.120756	172.16.8.8	172.16.8.49	SAMR	234	Connect5 response

▼ samr_UserInfo

Info

- Last Logon: Aug 8, 2026 21:08:26.686989900 Central Daylight Time
- Last Logoff: No time specified (0)
- Last Password Change: Aug 8, 2026 12:59:40.179085300 Central Daylight Time
- Acct Expiry: Infinity (absolute time)
- Allow Password Change: Aug 9, 2026 12:59:40.179085300 Central Daylight Time
- Force Password Change: Sep 19, 2026 12:59:40.179085300 Central Daylight Time

▼ Account Name

- Name Len: 12
- Name Size: 26
- Full Name: Raymond Vance

▼ Home Directory:

▼ Logon Script:

▼ Profile Path:

▼ Description:

▼ Workstations:

▼ Comments:

▼ Parameters:

▼ Lm Ovf Password:

Packets: 22473 · Displayed: 30 (0.1%)

Profile: Default

Figure 3. SAMR QueryUserInfo response identifies account rvance and full name Raymond Vance.

Network Behavior

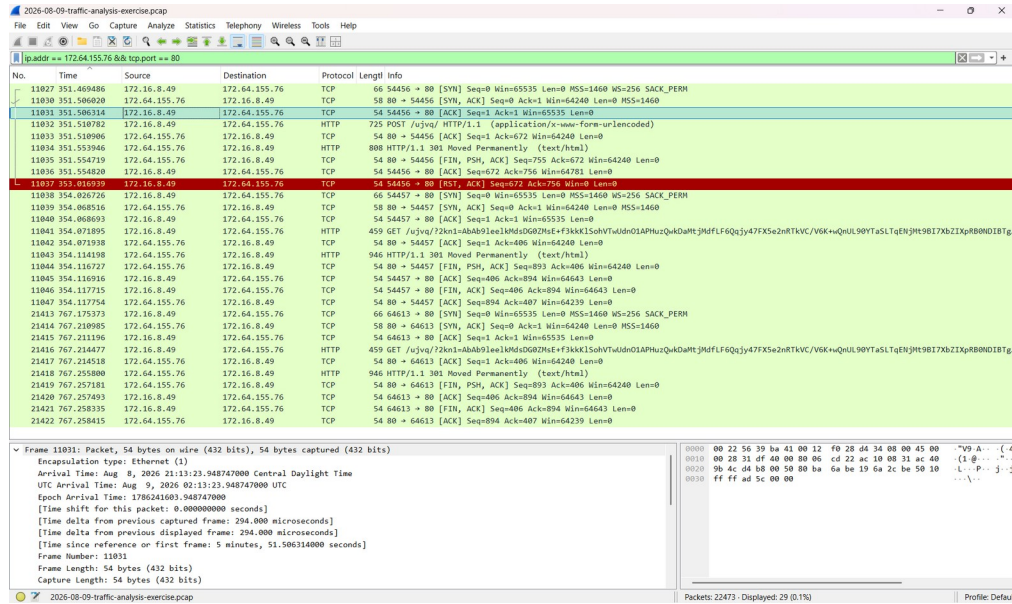


Figure 4. TCP connection establishment between 172.16.8.49 and 172.64.155.76.

The compromised client sent the initial SYN, received SYN-ACK, and completed the handshake with ACK. Relative capture times were 351.469486, 351.506020, and 351.506314. An HTTP request followed at 351.510782, establishing the order: connection initiation → TCP handshake → application-layer request.

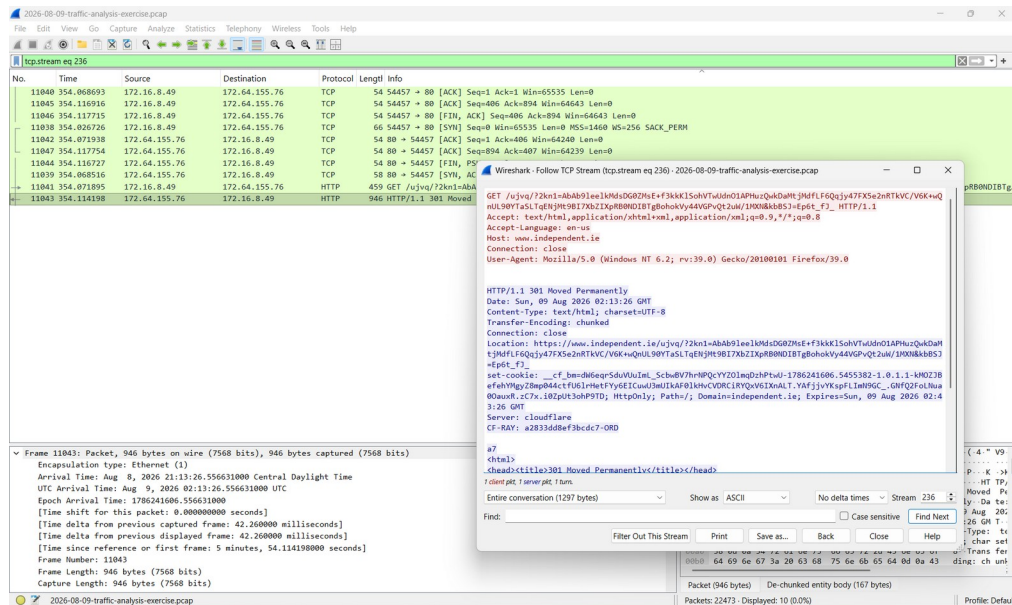


Figure 5. Follow TCP Stream view of HTTP GET request and 301 response.

The stream showed an HTTP GET to /ujvq/?... with Host header www.independent.ie and a browser-like Firefox user-agent. The server returned HTTP 301 Moved Permanently and redirected the URI to HTTPS. No immediate new TCP/443 connection attributable to that redirect was identified in the capture. The hostname is documented as an observed HTTP artifact; its appearance alone does not establish that the legitimate domain itself was malicious.

HTTP POST Activity

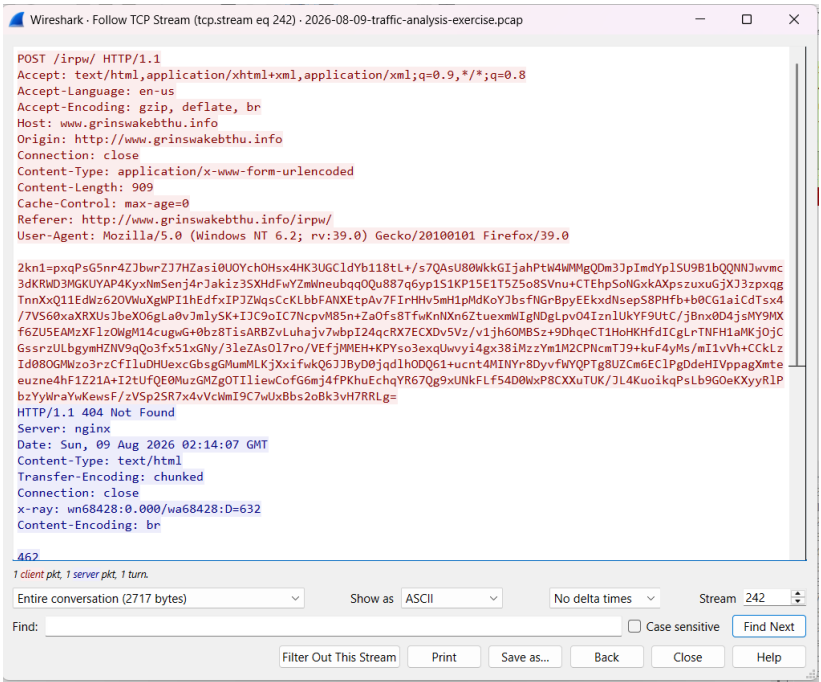


Figure 6. HTTP POST to /irpw/ with a 909-byte form-urlencoded body and subsequent 404 response.

A second alert-associated connection to 146.59.71.167 contained POST /irpw/ HTTP/1.1 with Host header www.grinswakebthu.info. The request declared Content-Type: application/x-www-form-urlencoded and Content-Length: 909. Its body began with the parameter 2kn1= followed by a long encoded-looking value. The server responded HTTP 404 Not Found. Because the body was not decoded and no host forensics were performed, the underlying data and whether it represented exfiltrated information were not determined.

Incident Timeline

Stage	Evidence / observation
Alert triage	FormBook C2 alerts supplied suspicious external IPs for investigation.
Endpoint correlation	172.16.8.49 communicated with both alert-associated external IPs.
Endpoint attribution	NBNS, Kerberos, and SAMR evidence identified DESKTOP-5NLV63K and rvance / Raymond Vance.
C2-associated HTTP activity	Client initiated TCP connection and sent GET /ujvq/?...; server returned 301 redirect.
Additional suspicious traffic	Client connected to 146.59.71.167 and sent POST /irpw/ with a 909-byte encoded-looking form value; server returned 404.

Risk Assessment & Recommended Response

Observed behavior is consistent with a compromised endpoint communicating with infrastructure identified by the exercise alerts as FormBook C2. Key risks include unauthorized external communications, possible exposure of endpoint or user data, continued malware activity, and potential credential or follow-on compromise. The PCAP does not establish the complete host impact or contents of the encoded POST body.

- Isolate the affected workstation and preserve relevant evidence before remediation.
- Block or monitor confirmed alert-associated indicators according to organizational policy, and hunt for the same indicators across other systems and logs.
- Perform endpoint investigation for malware execution and persistence; remove or reimagine the system as appropriate.
- Review authentication activity for the affected account and follow credential-reset procedures if compromise is established or suspected.
- Continue monitoring for recurrence and document containment, eradication, and recovery actions.

Skills Demonstrated

Wireshark packet analysis; TCP/IP and three-way handshake analysis; HTTP GET/POST inspection; NBNS, Kerberos, SMB/SAMR correlation; indicator identification; incident timeline reconstruction; evidence-based reporting; risk and remediation recommendations.

Limitations

This was a controlled training exercise using a supplied PCAP. Conclusions are limited to evidence visible in the capture. No malware was executed, no payloads were exported, and no endpoint memory, disk, EDR, or Windows Event Log evidence was analyzed. Encoded-looking request data was not decoded.